

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - III

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Class :SYIT	Batch : 01
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PART – A: Theory

Python provides built-in modules to work with **date**, **time**, and **calendar**. These modules help programmers display the current date, convert timestamps into readable dates, generate calendars, and compare dates. They are widely used in attendance systems, scheduling applications, event management, and report generation.

1. Working with Date and Time (`datetime` Module)

The `datetime` module provides classes to work with dates and time. It allows us to get the current date and time, format them, and perform calculations.

Syntax:

```
from datetime import datetime
today = datetime.now()
```

Explanation:

- `from datetime import datetime` imports the `datetime` class.
- `datetime.now()` returns the current system date and time.

2. Working with Timestamps (`time` Module)

A timestamp is the number of seconds elapsed since **January 1, 1970 (Unix Epoch)**. The `time` module converts timestamps into a readable date and time.

Syntax:

```
import time
time.ctime(timestamp)
```

Explanation:

- `ctime()` converts a timestamp into a human-readable date and time.

3. Displaying a Calendar (`calendar` Module)

The `calendar` module is used to display monthly and yearly calendars in Python.

Syntax:

```
import calendar
calendar.month(year, month)
```

Explanation:

- `month()` displays the calendar for a given month and year.
- `calendar()` displays the complete calendar for a year.

4. Comparing Two Dates

Python allows comparison of dates using comparison operators or by calculating the difference between dates.

Syntax:

```
date1 > date2
date2 - date1
```

Explanation:

- `>` checks which date is later.
- `-` returns the difference between two dates.

PART – B Questions:

a. Write a python code to print current date in different format.

INPUT:

```
from datetime import datetime
d = datetime.now()
print(d.strftime("%d-%m-%Y"))
print(d.strftime("%d/%m/%Y"))
print(d.strftime("%B %d %Y"))
print("S006 Sejal Deshmukh")
```

OUTPUT:

```
dsprac3.py ===
29-07-2026
29/07/2026
July 29 2026
S006 Sejal Deshmukh
>>|
```

b. Write a python code to convert time stamp to date stamp.

INPUT:

```
import time
t = time.time()
print(t)
print(time.ctime(t))

print("S006 Sejal Deshmukh")
```

OUTPUT:

```
prac3.py
1785314434.817265
Wed Jul 29 14:10:34 2026
S006 Sejal Deshmukh
```

c. Write a python code to develop calendar module.

INPUT:

```
import calendar
print(calendar.month(2026, 7))

print("S006 Sejal Deshmukh")
```

OUTPUT:

```
= RESTART: C:/Users/it11a
prac3.py
      July 2026
Mo Tu We Th Fr Sa Su
      1  2  3  4  5
 6  7  8  9 10 11 12
13 14 15 16 17 18 19
20 21 22 23 24 25 26
27 28 29 30 31

S006 Sejal Deshmukh
```

d. Write a python code to compare two dates.

INPUT:

```
from datetime import date
d1 = date(2026, 7, 29)
d2 = date(2026, 9, 14)
print(d1 < d2)
print(d2 - d1)

print("S006 Sejal Deshmukh")
```

OUTPUT:

```
prac3.py
True
47 days, 0:00:00
S006 Sejal Deshmukh
```

Curiosity Question:

1. **Why do computers use timestamps instead of storing dates directly?**
2. **How does Python know the current date and time?**
3. **Can Python display the calendar for any year, even 100 years in the future?**
4. **Why is comparing dates important in real-world applications?**
5. **What happens if you compare two dates from different years?**